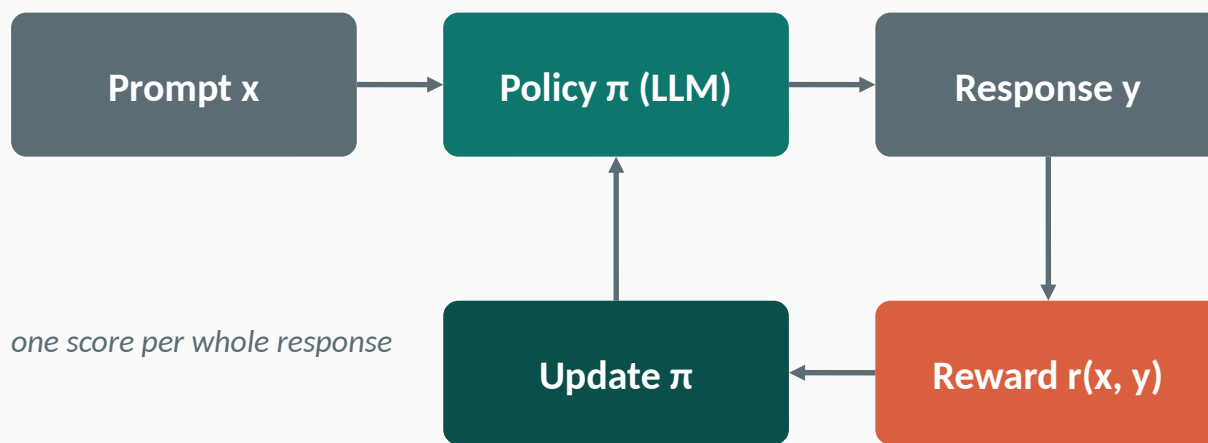




Policy π

The LLM itself: given a prompt, a probability distribution over complete responses.



Reference policy π_{ref}

A frozen copy of the model after supervised fine-tuning (SFT). The anchor we must not drift from.



Reward model r

A network trained on preference pairs (chosen vs rejected) to give one score per response.



KL penalty, weight β

How far π may move from π_{ref} . Small β : more freedom, more risk of reward hacking.

The objective (almost every method)

$$\max E[r(x, y)] - \beta \cdot \text{KL}(\pi \parallel \pi_{\text{ref}})$$

E[r]: give answers the reward model likes

$\beta \cdot \text{KL}$: ...without drifting far from the SFT model

The objective is the question: “which policy π gives the highest value of $E[r] - \beta \cdot \text{KL}(\pi \parallel \pi_{\text{ref}})$?”